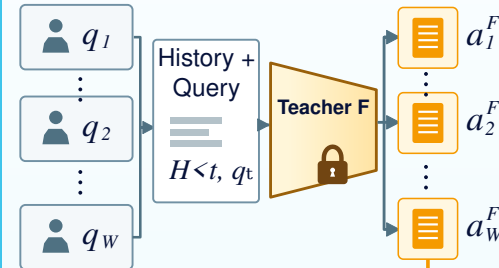
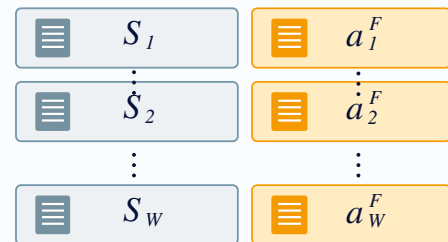


1 Warm-Start

Early dialogue turns
(1 ... W)



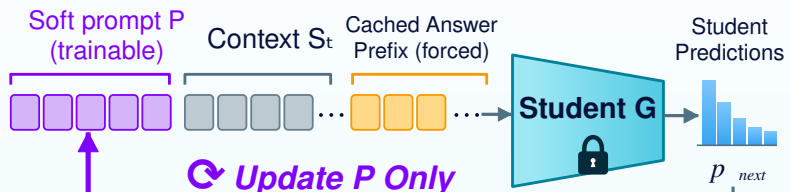
Warm-Start Pairs



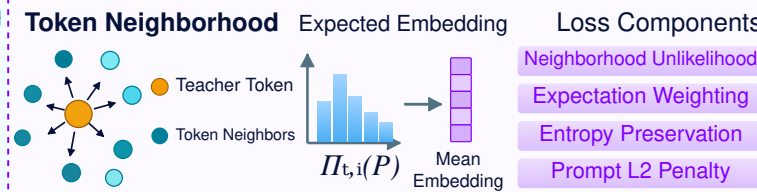
Cache Teacher Replies

2 Soft-Prompt Mining

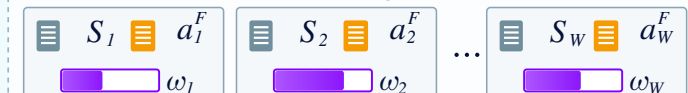
Input assembly (teacher-forced)



Mining Objective (teacher-forced student predictions vs. cached targets)



All Warm-Start Turns · Weights $\omega_t \propto r_t$

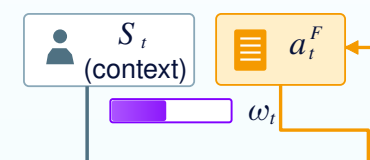


Optimize P · Freeze G · Anchoring Scores r_t

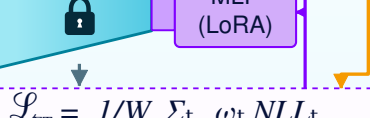
Discard After Mining

3 Local LoRA Distillation

All W warm-start turns
(anchoring weights $\omega_t \propto r_t$)



Student G (backbone)



$$\mathcal{L}_{FT} = 1/W \sum_t \omega_t NLL_t$$

$$NLL_t = -\log G_{\theta}(a_t^F | S_t)$$

Update LoRA Only

Adapted Student G_{θ}

One-Time Adaptation

4 Gated Serving

Acceptance Checks

- Response Fidelity (acceptance batch vs. F)
- Query Locality (vs. session centroid)

Both Pass?

Yes

No

Recheck

Teacher F Full history

Rollback

m_{cons} consecutive violations

Refresh Summary + Centroid

Updated context

Drift Monitor

Sliding window

Recent queries

Reply

Summary + Recent K Turns + Query

Prompt-Free Serving · No Online LoRA Updates

Session timeline



Frozen



Offline update



Rollback